

BHUBANESHWAR, India

WMO: 429710

Lat: 20.244N

Lon: 85.818E

Elev: 138

StdP: 14.62

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	56.9	58.9	46.0	46.1	77.1	48.9	51.5	75.1	13.4	74.7	11.4	74.2	2.1	0	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.8	102.7	81.1	100.4	81.0	98.2	80.4	85.4	94.7	84.6	93.3	84.0	92.1	10.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.5	176.3	89.4	82.8	171.7	88.5	82.3	169.3	88.0	50.2	94.5	49.3	94.1	48.5	92.0	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.4	16.1	DB	53.1	109.6	1.8	2.9	51.8	111.7	50.7	113.4	49.7	115.0	48.4	117.2	(4)
(5)			WB	50.8	87.6	2.0	1.6	49.4	88.8	48.2	89.7	47.0	90.6	45.6	91.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	72.1	78.4	84.8	88.4	89.9	87.2	83.9	83.6	83.5	81.7	76.8	71.9	(6)	
	DBStd	6.53	3.47	4.14	3.14	3.34	3.61	4.37	2.61	2.18	2.30	2.51	3.13	3.10	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11625	686	794	1078	1151	1236	1116	1050	1042	1005	982	805	680	(10)	
	CDD65	6151	221	374	613	701	771	666	585	577	555	517	355	216	(11)	
	CDH74	69165	1530	3293	6831	9066	10721	8781	6831	6569	6227	5173	2740	1403	(12)	
	CDH80	31535	397	1457	3389	5006	6408	4619	2614	2368	2261	1811	889	316	(13)	
(14) Wind	WSAvg	6.1	3.7	4.9	7.2	10.9	10.6	7.7	6.6	6.0	4.9	3.7	3.6	3.1	(14)	
(15) Precipitation	PrecAvg	59.6	0.6	0.8	1.1	1.3	2.6	7.8	14.7	14.4	9.4	6.3	1.3	0.1	(15)	
	PrecMax	76.0	3.5	3.3	3.9	5.9	10.3	19.4	31.5	29.4	18.3	21.0	7.1	1.9	(16)	
	PrecMin	35.6	0.0	0.0	0.0	0.0	0.0	1.0	7.1	4.3	0.0	1.1	0.0	0.0	(17)	
	PrecStd	11.0	0.9	0.9	1.2	1.4	2.4	4.4	4.9	5.3	3.9	4.8	1.7	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	98.7	102.1	105.6	107.0	107.5	95.4	93.4	93.6	93.1	90.6	87.7	(19)	
		MCWB	67.4	69.6	74.8	79.6	81.0	83.5	83.4	83.1	82.6	80.5	74.6	70.1	(20)	
	2%	DB	86.4	94.3	98.6	101.9	102.6	101.1	93.1	91.5	92.0	91.4	88.7	85.4	(21)	
		MCWB	68.4	70.6	75.1	79.8	83.2	83.1	83.0	82.8	82.2	79.4	73.5	68.6	(22)	
	5%	DB	84.2	91.4	96.3	99.3	100.3	97.5	91.2	90.0	90.7	89.8	87.1	83.5	(23)	
		MCWB	67.7	71.1	75.7	80.1	82.9	83.1	82.6	82.3	81.9	78.9	73.3	67.5	(24)	
	10%	DB	82.0	88.5	93.9	96.9	98.0	94.6	89.3	88.4	89.0	88.2	85.3	81.6	(25)	
		MCWB	67.0	70.5	75.8	80.0	82.4	83.0	82.3	82.0	81.6	78.4	72.6	66.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	78.3	82.8	85.5	86.9	86.8	85.1	84.6	84.5	83.4	79.6	74.8	(27)	
		MCDB	82.3	86.9	92.6	96.6	98.7	96.9	91.8	90.1	91.1	89.4	86.0	82.8	(28)	
	2%	WB	72.5	76.9	80.8	83.7	85.4	85.4	84.1	83.7	83.5	82.0	78.0	72.5	(29)	
		MCDB	79.8	84.4	90.9	95.8	96.0	94.0	90.1	89.0	89.6	87.6	83.8	79.4	(30)	
	5%	WB	71.0	75.5	79.8	82.6	84.7	84.5	83.4	83.0	82.7	81.2	76.7	70.6	(31)	
		MCDB	78.7	82.6	88.7	93.2	94.6	92.7	88.8	88.0	88.3	86.5	82.3	77.7	(32)	
	10%	WB	69.6	74.3	78.9	81.8	84.0	83.8	82.7	82.4	82.0	80.3	75.5	68.9	(33)	
		MCDB	77.3	81.3	86.5	91.1	93.3	91.3	87.5	87.1	87.0	85.3	81.2	77.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	22.2	20.8	19.4	17.8	13.5	9.7	9.6	10.4	13.1	17.7	21.0	(35)	
		MCDBR	23.8	25.1	23.3	21.9	20.2	18.7	12.9	11.8	13.2	15.3	19.1	22.2	(36)	
	5% WB	MCWBR	7.8	6.9	6.4	5.9	5.9	5.8	5.2	5.0	5.3	5.8	6.5	7.7	(37)	
		MCWBR	19.3	19.8	19.4	19.1	18.0	15.7	11.4	10.8	11.6	12.4	14.6	18.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.546	0.537	0.605	0.606	0.650	0.683	0.596	0.565	0.567	0.569	0.575	0.578	(40)	
		taud	1.843	1.896	1.785	1.830	1.768	1.707	1.908	2.004	1.991	1.938	1.840	1.790	(41)	
	Ebn at Noon		229	242	232	233	221	212	232	240	238	231	219	214	(42)	
		Edh at Noon	61	61	71	69	72	76	62	57	57	58	61	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1370	1647	1864	2048	1997	1508	1253	1318	1398	1464	1377	1262	(44)	
	RadStd		55	71	78	75	91	143	125	103	107	135	83	93	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.31	N/A	N/A	+1.76	+0.72	+0.54	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	+0.72	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page